

**Datapath Signal
Table**

Control Unit

Register File

Data Memory

Chooses input
between Read data 2
and offset towards
ALU's 2nd Input

For Branching
Instruction

Chooses Data transfer
from ALU or Data
Memory to Register File
for writing

ALU Control

Instruction Types (examples)

RegWrite

MemRead

MemWrite

ALUSrc

Branch

MemtoReg

ALUOp

R-type: add, sub, and, or etc.

1

0

0

0

0

0

10

Load: lw

1

1

0

1

0

1

00 (ADD)

Store: sw

0

0

1

1

0

X

00 (ADD)

Branch: beq

0

0

0

0

1

X

01 (SUB)

For instructions other than the ones given above, use your inference. Check what components of the Datapath are being used and what read/write are needed.